Paul Parisot

Software Engineer - Embedded Linux & C++ Systems

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About

Systems software engineer specialized in high-performance C++ development and low-level Linux environments.
Deep understanding of runtime memory layouts, custom allocators, asynchronous networking architectures, and hardware constraints.
Proactive debugger proficient with defensive programming, strict memory profiling (Valgrind/GDB), and industrial TDD practices.

Skills

Core Systems: C, C++ (C++14/17/20), Python scripting, Bash, x86/ARM Assembly stubs
Embedded Linux: Memory Management (`heap, stack, mmap, shm`), POSIX threads, System Services, Driver communication boundaries
Tooling & Toolchains: CMake, GCC, GDB, Valgrind (Memcheck), Git, Automated Cross-Compilation
Networking & Quality: TCP/IP paradigms, Async I/O (`Boost.Asio`), TDD (Test-Driven Development), Code Auditing

Professional Experience

Faurecia / Forvia - Software Engineer (Embedded C++ / Systems)

- Developed and optimized robust low-level C++ software modules for real-time automotive telemetry systems.
- Managed strict resource constraints and memory-safety guarantees on target architectures.
- Enforced zero-regression policies through rigorous **TDD (Test-Driven Development)** and automated validation loops.
- Operated within an industrial Agile/Scrum environment, interfacing closely with system requirements via **Jira**.

Gno.land - Core Software Engineer (Distributed Runtime Engines)

- Refactored core runtime sub-systems, focusing on multi-threaded thread safety and resource-efficient concurrency.
- Isolated and resolved **6 critical execution bugs** (race conditions, memory leaks), committing over 10,000 lines of production code.
- Designed full integration test suites to validate complex system-level protocols and distributed communications.

Epitech / IONIS - Systems Programming Instructor & Code Auditor

- Instructed **400+ students** in advanced C/C++ memory management, low-level pointers, and Unix system calls.
- Conducted exhaustive code audits to eliminate undefined behaviors, memory corruption, and unhandled system signals.

Performance & Low-Level Technical Projects

rodox – Low-Latency Linux Systems Engine

Engineered a modular runtime architecture under Linux utilizing a data-oriented layout (ECS) to eliminate L1/L3 cache misses.
Developed custom **static pool allocators** to entirely bypass OS-level heap allocation latencies (`malloc/new`).
Implemented high-performance asynchronous networking and communication layers using `Boost.Asio` over TCP.
Profiled and hardened using **CMake**, **GDB**, and **Valgrind** to guarantee absolute deterministic execution.

Skills: C++, Boost.Asio, Memory Management, Linux System APIs, Profiling (Valgrind), CMake, Concurrency

cclass – Runtime Memory Mutation Engine

Designed a pure C framework that dynamically modifies compiled assembly instructions inside raw memory pages.
Utilized `mmap` system calls with precise protection flags to bypass traditional virtual table dispatch overheads.
Enforced strict memory alignment rules to maintain absolute stability across underlying hardware execution targets.

Skills: C, Linux Systems, Assembly, mmap, Memory Safety, Low-level Debugging

Sledge – Cross-Cloud Data Migration & Orchestration Engine

Co-authored a backend platform in Go optimized for transferring huge data streams securely across disparate networks.
Built optimized abstraction layers to process relational information schemas without operational overhead or stream interruption.

Skills: Go, Data Streaming, Network Robustness, System Automation

Education

Epitech, Paris 2022 – 2027
Master's Degree in Information Technology (Master's level diploma)
Level-7 Certification - Expert in Software Engineering
Specialization: Systems Architecture

McGill University, Montréal 2025 – 2026
Certificate in Management
Technical Leadership & Strategy